

Empirical Evaluation of Proximal Policy Optimization for UAV Navigation in Constrained Indoor Environments

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Abstract—Autonomous navigation of unmanned aerial vehicles (UAVs) in constrained indoor environments remains a challenging problem due to limited maneuvering space and high collision risk. This paper presents an empirical evaluation of a reinforcement learning-based approach for UAV path planning using Proximal Policy Optimization (PPO) in a high-fidelity AirSim simulation environment. The UAV operates in a cluttered warehouse setting with static obstacles, where it must reach predefined goal locations while minimizing collisions and trajectory inefficiencies. We conducted extensive experiments to analyze the reward function shaping, training dynamics, hyperparameter sensitivity, and navigation performance across multiple goal configurations. Results show that the PPO-based agent achieves an overall success rate of (78%) with stable convergence and improved path efficiency, while maintaining low collision rates in less constrained regions. However, performance degrades in obstacle-dense areas requiring sharp maneuvers, highlighting limitations in generalization. The study provides insights into the impact of reward design and environment complexity on reinforcement learning performance for indoor UAV navigation.

Index Terms—UAV navigation, reinforcement learning, artificial intelligence, Proximal Policy Optimization (PPO), path planning, indoor drone maneuvering, obstacle avoidance, autonomous robotic systems, embedded control.

I. INTRODUCTION

The growing adoption of unmanned aerial vehicles (UAVs) in industrial and logistics settings has increased the demand for reliable autonomous navigation in constrained indoor environments. Applications such as warehouse inventory management, inspection, and internal delivery require UAVs to operate safely within cluttered spaces characterized by narrow aisles, static obstacles, and limited maneuvering freedom. In such settings, path planning and collision avoidance become significantly more challenging than in open or outdoor environments.

Classical path planning methods, including graph-based and rule-based approaches, can provide optimal solutions under

structured assumptions, but they often struggle to adapt to complex, partially observable environments. These limitations have motivated the use of reinforcement learning (RL), where an agent learns navigation policies through interaction with the environment. RL-based methods are particularly suited for continuous control tasks, enabling UAVs to learn smooth and adaptive motion strategies without relying on predefined models of the environment.

Among reinforcement learning approaches, policy gradient methods have demonstrated strong performance in continuous control problems. Proximal Policy Optimization (PPO) in particular provides a favorable trade-off between training stability, sample efficiency, and implementation complexity. Unlike more recent off-policy methods such as Soft Actor-Critic (SAC) or Deep Deterministic Policy Gradient (DDPG), which often require careful tuning of multiple interacting components, PPO employs a clipped objective that constrains policy updates, resulting in more predictable and stable learning behavior. This stability is especially important in safety-critical tasks such as UAV navigation, where erratic policy updates can lead to unsafe actions. In addition, PPO's on-policy formulation simplifies integration with simulation environments and reduces sensitivity to hyperparameter configurations, making it a practical choice for real-time control scenarios. While PPO has been applied to aerial navigation in prior work, most studies focus on open or large-scale environments, with limited analysis of its performance in constrained indoor settings such as warehouses.

The main contributions of this paper are as follows:

- 1) An empirical evaluation of PPO for UAV path planning in a constrained warehouse-like environment using a high-fidelity simulation platform.
- 2) A structured analysis of reward function design and its

impact on navigation behavior, including goal-reaching efficiency and collision avoidance.

- 3) An evaluation of policy generalization across multiple goal locations, identifying performance degradation in obstacle-dense regions.

II. RELATED WORK

Unmanned aerial vehicle (UAV) path planning has been extensively studied using both classical optimization techniques and learning-based approaches. Traditional methods such as A* and Dijkstra's algorithm provide deterministic and optimal solutions in structured environments, but their effectiveness degrades in dynamic or partially observable settings due to limited adaptability [1]. These limitations have motivated the adoption of reinforcement learning (RL), which enables agents to learn navigation policies directly from interaction with the environment.

Recent advances in deep reinforcement learning (DRL) have shown strong potential for UAV navigation in complex environments. Surveys such as [2] highlight the transition from classical planning to learning-based methods, emphasizing RL's ability to handle uncertainty and high-dimensional state spaces. In simulated environments, DRL approaches have demonstrated the ability to learn collision-aware navigation policies without explicit modeling of the environment dynamics. For example, Yan *et al.* [3] applied DRL for real-time UAV path planning in AirSim, achieving effective obstacle avoidance. Similarly, Tu and Juang [4] demonstrated robust UAV navigation in 3D environments using RL-based control policies.

Among RL methods, policy gradient algorithms are particularly suited for continuous control tasks such as UAV flight. Proximal Policy Optimization (PPO), introduced by Schulman *et al.* [5], improves training stability through a clipped surrogate objective that limits destructive policy updates. PPO has been successfully applied to UAV navigation problems, including urban and cluttered environments. Chikhaoui *et al.* [6] demonstrated that PPO can effectively handle high-dimensional state spaces in UAV navigation tasks, while maintaining stable convergence behavior. Comparative studies, such as [7], further confirm PPO's reliability relative to other policy gradient methods.

Alternative RL algorithms have also been explored for UAV navigation. Deep Deterministic Policy Gradient (DDPG) [8] enables continuous control but is known to suffer from training instability and sensitivity to hyperparameters. Soft Actor-Critic (SAC) [9] improves sample efficiency through entropy maximization, but introduces additional computational complexity. While these methods offer advantages in specific scenarios, PPO remains a practical choice for single-agent navigation due to its balance between stability and implementation simplicity.

Despite these advances, most existing work focuses on open or large-scale environments, with limited attention to constrained indoor settings such as warehouses [4]–[6]. Such environments introduce additional challenges, including narrow navigation corridors, dense obstacle distributions, and sharp

maneuvering requirements. This work builds on prior studies by providing an empirical evaluation of PPO-based UAV navigation specifically in a constrained indoor environment, with emphasis on reward design, hyperparameter sensitivity, and environment-induced performance limitations.

III. METHODOLOGY AND EXPERIMENTAL SETUP

In this section, we describe the experimental framework used to evaluate reinforcement learning for UAV navigation in constrained indoor environments.

A. Simulation Environment

We use Microsoft AirSim (v1.8.1) to conduct our experiments. AirSim provides a high-fidelity physics-based simulation environment for aerial vehicles that is integrated with Unreal Engine (v4.27) for real-time 3D visualisation. Since we focus on dense indoor environments, we constructed a three-dimensional L-shaped warehouse of approximately 1300 m², populated with static obstacles such as storage racks and supporting columns for the UAV to operate within as shown in Figure 1. The layout introduces narrow corridors and constrained maneuvering regions, making collision avoidance a central challenge.

The UAV maintains a fixed altitude throughout training and testing, simplifying the navigation task to planar motion while preserving obstacle interactions. We enable perception through a combination of onboard sensors, including four directional distance sensors and a LiDAR unit, which provide information about nearby obstacles. These observations form the basis for decision-making under partial environmental awareness.

B. Reinforcement Learning Formulation

At each discrete timestep, the UAV receives a state representation, selects an action, and observes a scalar reward

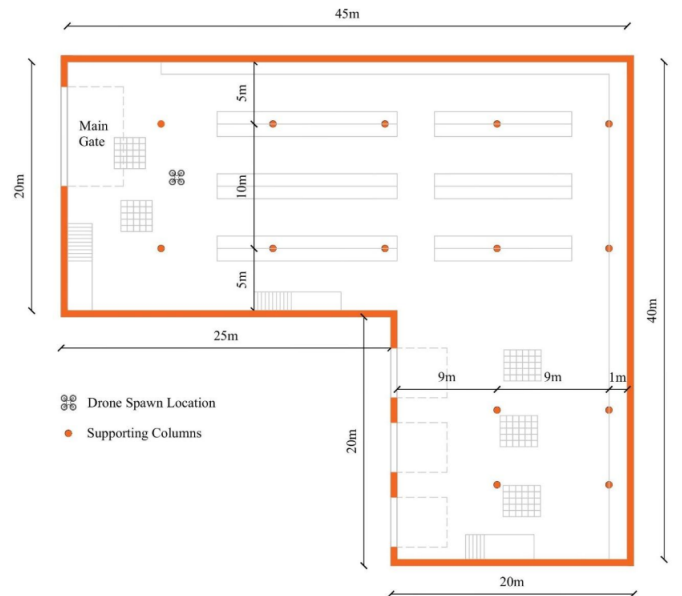


Fig. 1. Warehouse Layout and Geometry and Static Obstacle Placement

that reflects its performance. The state is represented as a 16-dimensional vector that captures both the UAV's kinematic properties and its spatial relationship to the goal and surrounding obstacles. This includes UAV position (3D), UAV velocity (3D), goal position (3D) relative distance to goal (3D), and proximity readings from onboard sensors (4 readings). Actions correspond to continuous velocity commands in three dimensions, bounded within a normalized range to ensure stable control.

Episodes terminate when the UAV successfully reaches the goal region, collides with an obstacle, or exceeds a predefined step limit. This formulation encourages the agent to learn efficient trajectories while respecting safety constraints.

C. PPO-Based Navigation Policy

We base the navigation policy on Proximal Policy Optimization (PPO), an actor-critic policy gradient method [5]. The actor outputs a Gaussian policy for continuous control, while the critic estimates state values. We compute advantage estimates using Generalized Advantage Estimation, and policy updates follow a clipped surrogate objective to ensure stable training.

D. Reward Function Design

We define a reward function that balances goal-directed behavior with safety and efficiency as follows:

$$R_t = R_{\text{goal}} + R_{\text{progress}} + R_{\text{prox}} + R_{\text{collision}} + R_{\text{near}} + R_{\text{step}} + R_{\text{timeout}} + R_{\text{smooth}} \quad (1)$$

Each component of the reward function targets a specific aspect of the navigation task. We assign a large terminal reward R_{goal} upon reaching the goal, while a dense progress term R_{progress} , defined by the reduction in distance to the target, guides and accelerates learning during early training. We complement this with a proximity-based shaping term R_{prox} that increases nonlinearly as the UAV approaches the goal, thereby improving convergence near the target.

We enforce safety through a fixed collision penalty $R_{\text{collision}}$ and a distance-dependent near-miss penalty that discourages operation close to obstacles. To promote efficiency, a small step penalty R_{step} and a timeout penalty R_{timeout} discourage unnecessarily long trajectories. In addition, a smoothness term R_{smooth} encourages consistency between consecutive actions, reducing abrupt control variations.

The combined effect of these components yields a balanced objective that promotes efficient and collision-aware navigation. We summarize the corresponding reward and penalty values in Table I, where P_{curr} and P_{goal} denote the current UAV position and target goal position, respectively. The terms D_{prev} and D_{curr} represent the previous and current distances to the goal, while $D_{\text{nearest_obs}}$ denotes the distance to the nearest obstacle and D_{safe} is set to 0.6 m.

We illustrate some of the rewards and penalties in Figure 2 where we show the penalty radius for hitting warehouse support beams as an example, and the rewards for reaching the proximity of a sample target.

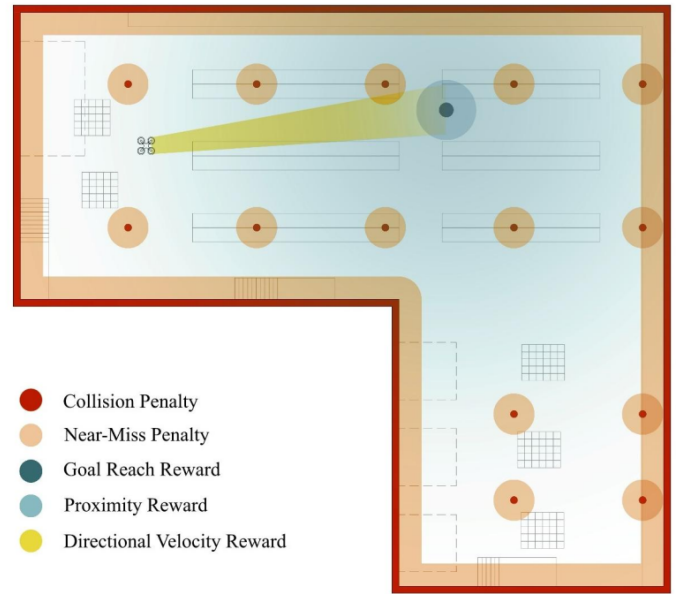


Fig. 2. Visual Presentation of the Rewards and Penalties

TABLE I
REWARD FUNCTION COMPONENTS

Term	Formulation	Value
R_{goal}	λ_g if $ P_{\text{curr}} - P_{\text{goal}} \leq 0.5$	$\lambda_g = 300$
R_{progress}	$\lambda_p \times (D_{\text{prev}} - D_{\text{curr}})$	$\lambda_p = 0.1$
R_{prox}	$\lambda_x = \min\left(\frac{10}{D_{\text{curr}} + 10^{-3}}, 20\right)$	—
$R_{\text{collision}}$	$-\lambda_c$	$\lambda_c = 10.0$
R_{near}	$-\lambda_n \times \left(1 - \frac{D_{\text{nearest_obs}}}{D_{\text{safe}}}\right)$	$\lambda_n = 0.05$
R_{step}	$-\lambda_s$	$\lambda_s = 0.1$
R_{timeout}	$-\lambda_t$ if $\text{Step}_{\text{current}} \geq 120$	$\lambda_t = 10.0$
R_{smooth}	$\lambda_m \times (1 - \text{Action}_{\text{curr}} - \text{Action}_{\text{prev}})$	$\lambda_m = 10.0$

E. Training Configuration

We train the agent over multiple episodes, each with a fixed maximum number of timesteps. During each episode, the UAV interacts with the environment, collects trajectories, and updates the policy using mini-batch gradient descent.

We then examine the sensitivity of the learning process to key hyperparameters through controlled experiments. Increasing the batch size from 64 to 128 produces smoother reward curves, reduces variance in policy updates, and significantly lowers collision rates. The larger batch size improves gradient estimation and leads to more stable convergence, resulting in substantially higher success rates compared to smaller batches. Similarly, we observe that an intermediate critic learning rate (e.g., 0.0003) provides the best trade-off between convergence speed and stability, whereas higher values introduce oscillations and lower values slow down learning.

During training, we periodically save model checkpoints as PyTorch .pth files. Each checkpoint stores the parameters of the actor and critic networks at a specific training stage. We later evaluate these checkpoints on the test scenarios

TABLE II
PARAMETER TUNING AND FINAL PARAMETERS CONFIGURATION

Parameter	Value
Maximum Training Episodes	700
Max Steps per Episode	120
Actor Learning Rate	0.0001
Critic Learning Rate	0.0003
Discount Factor (γ)	0.95
GAE (λ) Advantage smoothing	0.90
Batch Size	128
PPO Clipping parameter	0.2
Entropy Coefficient	0.01
Gradient Clipping	Applied

to analyze learning progression and to identify the best-performing policy.

The results also highlight the influence of other parameters. The discount factor γ governs the balance between short-term and long-term rewards, while the GAE parameter λ controls the bias-variance trade-off in advantage estimation. Entropy regularization promotes exploration during early training, but we reduce its effect over time to enable convergence, as excessive entropy leads to unstable behavior and insufficient entropy causes premature convergence. Based on these observations, we select a set of final training parameters that provide stable and effective learning. We summarize these parameters in Table II. Due to the lack of space, we only present the final values used in this study rather than the ablation study results.

F. Evaluation Protocol

To assess generalization, we evaluate the trained policy on a set of previously unseen goal locations distributed across the warehouse environment. For each goal, we conduct multiple evaluation episodes from a fixed initial position. During testing, exploration noise is disabled so that the learned policy operates deterministically.

We measure performance using three metrics. The success rate (SR) is the proportion of episodes in which the UAV reaches the goal within the maximum allowed steps N_{EGR} to the total episodes N_T :

$$SR = \frac{N_{EGR}}{N_T} \times 100\% \quad (2)$$

The collision rate (CR) represents the fraction of episodes that terminate due to collisions with obstacles N_C to the total episodes N_T :

$$CR = \frac{N_C}{N_T} \times 100\% \quad (3)$$

We define path efficiency (PE) as the ratio between the number of steps taken to reach the goal N_S and the maximum allowed steps Max_S , with lower values indicating more efficient trajectories:

$$PE = \left(1 - \frac{N_S}{Max_S}\right) \times 100\% \quad (4)$$

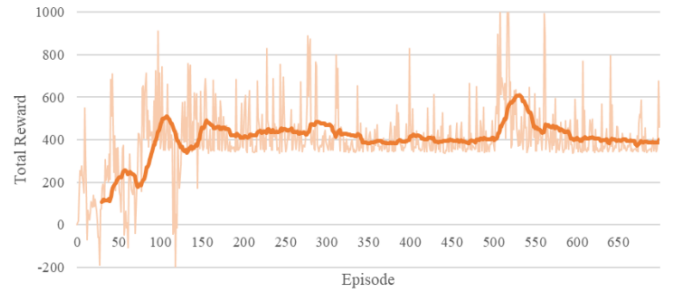


Fig. 3. Total Reward in Goal Training

We average all metrics over multiple runs to ensure consistency. Together, these measures provide a comprehensive assessment of navigation performance across different goal locations and levels of environmental complexity.

IV. RESULTS AND DISCUSSION

This section presents the experimental results obtained from training and evaluating the PPO-based UAV navigation agent in the constrained warehouse environment. The analysis focuses on two aspects: (i) learning dynamics during training, (ii) task-level performance in terms of success, safety, and efficiency over nine unseen goal locations.

A. Learning Dynamics During Training

We analyze the learning dynamics of the PPO agent to understand how training evolves and which factors most influence performance. Figure 3 shows the evolution of cumulative episode rewards. The reward increases rapidly during the early stages of training, indicating that the agent quickly acquires basic goal-directed behavior. This initial phase is followed by a stabilization regime in which rewards fluctuate within a bounded range, reflecting incremental policy refinement. Convergence is typically observed within the first 80-100 episodes.

Figure 4 provides further insight into training stability by showing the evolution of actor loss, critic loss, and collision counts. The critic loss initially exhibits significant oscillations due to unstable value estimation under sparse and shaped rewards. As training progresses, both actor and critic losses stabilize, indicating improved consistency in policy updates. In parallel, collision counts decrease steadily, confirming that the agent learns to reduce unsafe interactions over time. However, collisions do not vanish entirely, suggesting that the learned policy does not fully resolve safety constraints in complex regions.

Overall, the learning dynamics demonstrate that PPO can achieve stable convergence in the constrained warehouse environment, but the final performance remains sensitive to reward design and hyperparameter selection.

B. Task-Level Performance on Unseen Goals

To evaluate task-level performance, we test the trained policy on nine previously unseen goal locations distributed across the warehouse environment (Fig. 5). We selected these

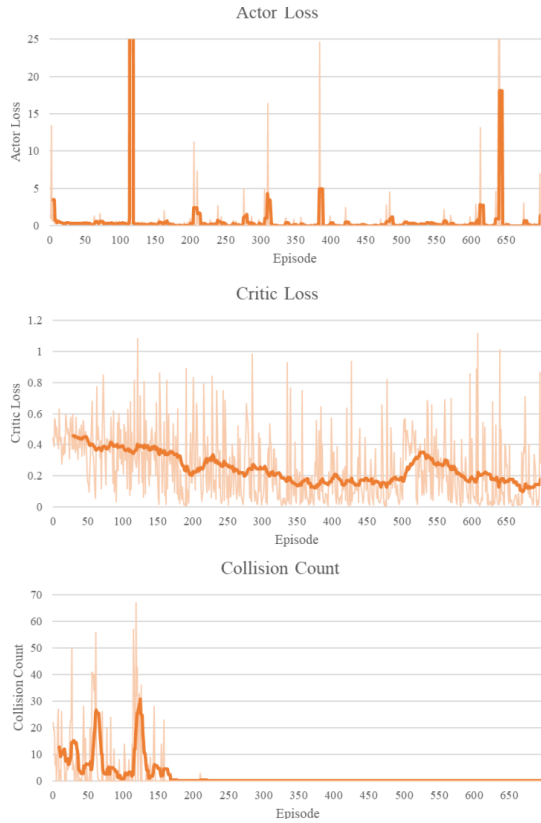


Fig. 4. Learning Dynamics: Actor and Critic, and Collision Rates

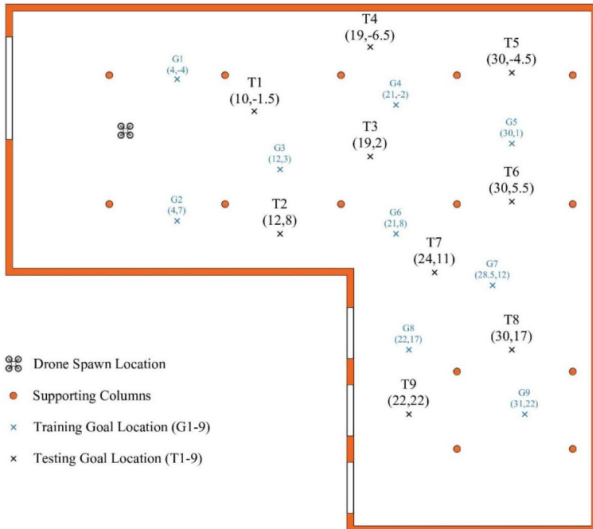


Fig. 5. The Location of Nine Previously Unseen Test Goals

locations to cover both central and peripheral regions with varying levels of spatial complexity.

In Figures 6, 7, 8, we summarize the results for success rate (SR), collision rate (CR), and path efficiency across all nine test goals. Across all evaluated checkpoints, evaluation runs and all goal locations, the agent achieves an overall success rate of approximately 78% indicating that it can reliably learn

goal-directed navigation in most scenarios. Collision rates remain relatively high with an average of 22% across all targets and all checkpoints confirming that obstacle avoidance remains a key limitation. The agent often prioritizes reaching the goal over maintaining safe distances from obstacles, reflecting the trade-off imposed by the reward structure.

Path efficiency stabilizes within a moderate range (approximately 50-70%), indicating that once a feasible trajectory is identified, the agent follows reasonably direct paths. Nevertheless, efficiency degrades in cluttered areas, where detours and hesitation increase trajectory length.

A more detailed analysis reveals a strong dependence on spatial complexity. Goals located along direct and unobstructed paths (e.g., T1 and T3) achieve near-perfect success rates with low collision rates and high efficiency. In contrast, goals that require sharp turns or navigation through narrow passages (e.g., T4 and T9) exhibit lower success rates, higher collision frequencies, and reduced efficiency. Intermediate cases (e.g., T2, T5, and T6) show moderate performance with increased variability.

These results indicate that the learned policy generalizes well to regions similar to those encountered during training but struggles in areas requiring complex maneuvering. In particular, locations outside the effective spatial coverage of training trajectories show degraded performance, highlighting limitations in the policy's ability to extrapolate beyond the training distribution.

C. Discussion

All in all, three key limitations emerge:

- 1) **Incomplete collision avoidance**: Despite improvements during training, collision rates remain higher than desired, indicating insufficient safety awareness. This is especially true in late-stage collisions, where the agent approaches the goal but fails to execute precise final maneuvers.
- 2) **Limited generalization**: Performance degrades in regions not well covered during training, particularly those requiring sharp turns or complex maneuvers. As the agent could deviate significantly from efficient paths due to uncertainty in less-visited regions.
- 3) **Sensitivity to training design**: Hyperparameter choices and reward structure significantly influence convergence and final performance.

These findings suggest that while PPO provides a strong baseline, further improvements in reward shaping, state representation, and training diversity are required to achieve robust and scalable UAV navigation in real-world environments.

V. CONCLUSION

This paper presented an empirical evaluation of a Proximal Policy Optimization (PPO)-based approach for UAV navigation in a constrained indoor warehouse environment. Using a high-fidelity AirSim simulation, we examined how reward design, hyperparameter configuration, and environment structure influence learning behavior and navigation performance. The

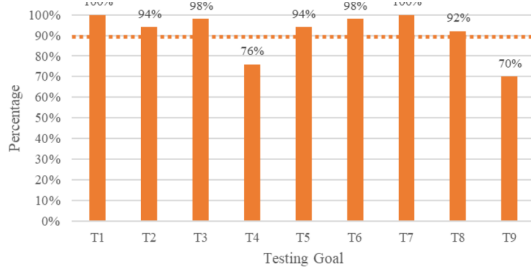


Fig. 6. Success Rate Across Nine Previously Unseen Target Locations (Results Reported from best checkpoint for each target)

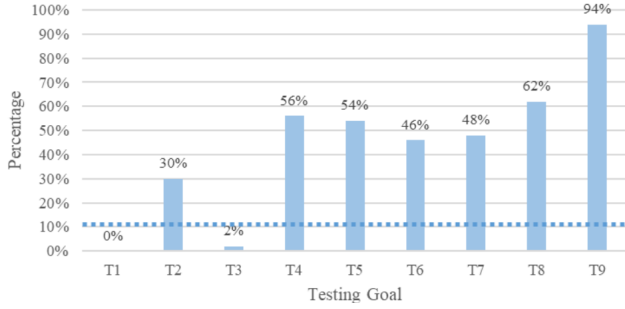


Fig. 7. Collision Rate Before Reaching Nine Previously Unseen Targets (Results Reported from best checkpoint for each target)

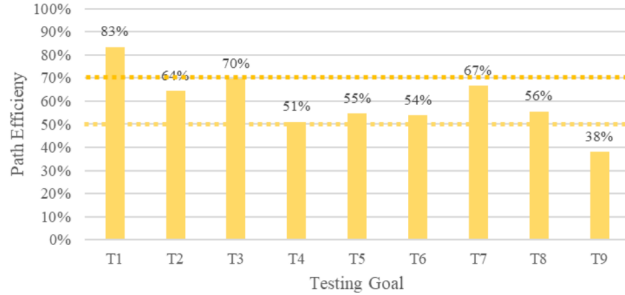


Fig. 8. Path Efficiency for Nine Previously Unseen Targets (Results Reported from best checkpoint for each target)

results show that PPO can learn stable policies that achieve reliable goal-reaching behavior and moderate path efficiency in structured indoor settings.

However, the analysis also reveals clear limitations. Collision avoidance remains inconsistent, particularly in narrow corridors and obstacle-dense regions. In addition, performance varies significantly across goal locations, indicating limited generalization beyond areas well represented during training. The results further confirm that training outcomes are sensitive to hyperparameter choices, especially batch size and learning rates, which directly affect stability and convergence.

Future work should focus on improving both robustness and generalization. This includes refining reward design to better balance safety and goal-reaching objectives, enhancing state representation through richer perception (e.g., depth or semantic information), and increasing training diversity to cover a wider range of spatial configurations. In addition, hybrid approaches that combine reinforcement learning with classical planning or hierarchical control may help address

precise maneuvering in complex regions. Extending the framework to dynamic environments and validating performance on real UAV platforms are also important steps toward practical deployment.

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